

Team: Citrix Are For Kids!

Project Title: Study Together

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Product Requirements Design 1.0

Introduction

Problem Statement

Throughout their high school and university experiences, students spend much time collaborating with peers, teachers, and TAs, as well as participating in group projects, study groups, and review sessions. However for many students, physically getting together in a group setting can be a challenge. There are many non-traditional students such as those who work full or part-time, parents, and disabled people, who find it difficult to set aside the time to travel to another location for a group meeting when they have many other priorities. Additionally, for students, finding a physical meeting space on campus can be difficult, especially during high traffic times such as midterms and finals. There are also often groups of students from various schools and courses who get together to study for exams such as the SAT, GRE, and MCAT, who would benefit greatly from an online platform that allows them to create ongoing study groups while practicing for these exams. Currently, the best solutions for remote collaboration and studying are a combination of Skype or Google Hangouts and Google Drive, but these lack integration with one another and are also not focused on student study groups.

Project Outcome & Innovations

The aim of this project is to make studying with peers significantly easier. Study Together is a platform that combines video/audio meetings, screen sharing, and document editing to make group collaboration possible without being physically near the group members. It also provides a matchmaking service for students looking to join a study group; the application displays available study sessions for specific courses and allows students to request to join an existing study session if they find one that suits their needs. These features not only improve studying with a group when people are unable to meet in person, but they make conventional, in-person study meetings obsolete by providing invaluable study tools and built-in smarts.

Study Together allows students to enter their school and courses for the term. Based on this information, the application creates a unique dashboard for the user that displays a list of their classes

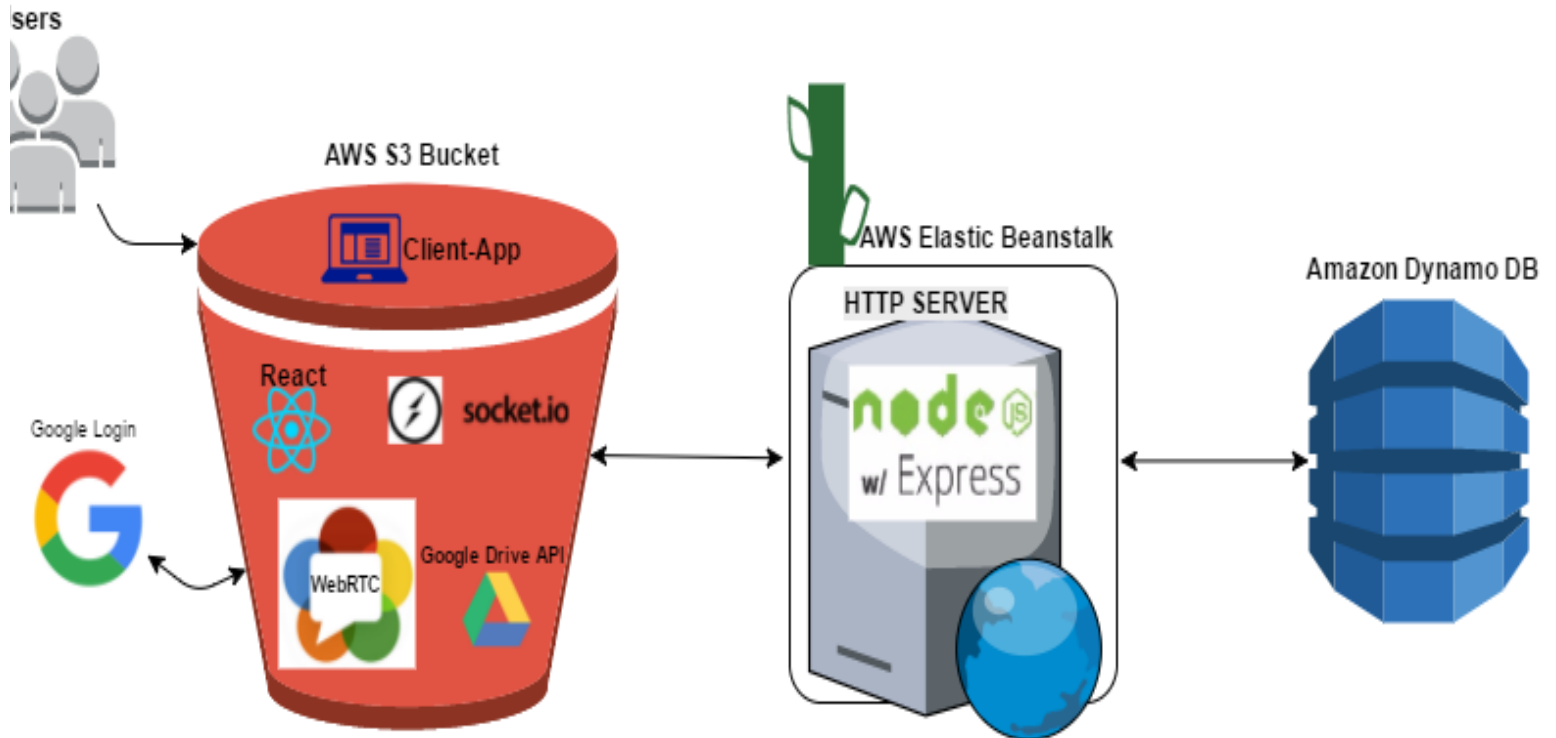
in the center. The user is able to view each class's page, where all current study groups will be displayed as well as file uploads from classmates; these files may be lecture notes, notes from a review session, a photo of the blackboard during lecture, or any other useful information that a student thinks could be helpful for his or her classmates. From this page, a user is able to start a study session and make it public to the whole class so that anyone wanting to join a study group can simply navigate to the class's page, find the group, and request to join. The aim of this matchmaking feature is to make finding a study group simple and intuitive.

The main feature of Study Together is the creation of study groups. From the chat sidebar, users are then able to create study groups and invite specific people if they wish to keep their group private. Invited members of the group are also able to invite additional members to the group if they wish to do so. This creates an additional chat, specific for the members in the group. These sessions are not viewable from a specific class's page in the "Ongoing Sessions" section of a class page unless explicitly declared public by the session's creator. In a study group session, users are able to use audio and video to speak to each other, as well as text chat to communicate additional information such as web pages, images, and files with other group members. Screen sharing is yet another capability enabled by the application that makes sharing information with others simple. The study group session is also connected to Google Drive to cohesively integrate collaborative document editing in the application. This way, all the group members can work on a single Google Doc, Sheet, or Presentation together while speaking with one another and reviewing each other's work. To make remote collaboration even better, Study Together has built-in smarts that automatically bring up web pages, articles, and images relevant to the topic being discussed in the group meeting.

The application also contains a chat feature, which is collapsible from the side of the screen. The chat bar is separated into two sections: one for the user's classes, where a list of all their classmates is displayed, and one for all the user's contacts, which may be previous classmates or friends from other schools. Users can also add contacts by simply searching for the person. The chat feature is separate from the video session, allowing users to coordinate meetups, discuss topics at hand, and preserve history of communication. The application also provides a chat transcript of the messages sent between users once the session is completed, which can be used as a summary of the whole group study session. If so desired, the file can then be uploaded to a specific course's upload section for the whole class to access and review for an upcoming exam or project.

Study Together's most prominent innovation is providing multiple forms of communication that students currently use separately all into one, cohesive application as well as providing an interface for students in the same courses to share relevant and helpful information with their peers. It allows an entire class of students to study with one another without the hassle of finding a physical meeting space and the communication difficulties that come with studying with a large group.

System Architecture



Requirements

User Stories

1. First Time User Login
 - a. As a student, I create an account so I can begin using the application.
 - b. Acceptance Criteria
 - i. User is logged into account
 1. User ID in database
 - ii. Can generate and refresh an auth token
 - iii. User is led to user management page (where he/she enters school info, class info, user info)
 - c. Test Cases: <https://github.com/peterwerner/citrix-capstone/blob/master/server-app/test/test.js>
2. Existing User Login
 - a. As a student, I can log into my account so I can study with my peers.
 - b. Acceptance Criteria
 - i. User is logged into account
 1. User ID in database
 - ii. Can generate and refresh an auth token
 - iii. Can view and interact with dashboard
 - c. Test Cases: <https://github.com/peterwerner/citrix-capstone/blob/master/server-app/test/test.js>
3. Logout
 - a. As a student, I can logout of my account when I am done using the application.
 - b. Acceptance Criteria
 - i. User cannot access any of his/her courses or study sessions
 - ii. User cannot access his/her dashboard
 - iii. User is led directly to login page
 - c. Test Cases: <https://github.com/peterwerner/citrix-capstone/blob/master/client-app/src/test/App.test.js>
4. Create a Study Session
 - a. As a student, I can create a study session so I can begin a study group.
 - b. Acceptance Criteria
 - i. New video session is created with a group key, that is stored in the database
 - ii. Creator can specify the class, project, test, topic, etc. that the group is studying for (since there can be multiple subjects/exams to study for a certain class)
 - iii. Creator is automatically added to the video session
 - iv. Video, audio, and text chat are automatically started
 - c. Test Cases: <https://github.com/peterwerner/citrix-capstone/blob/master/server-app/test/test.js>

5. Invite Users to a Study Session
 - a. As the member of a study session, I can invite additional users to join my study group so I can study with more people.
 - b. Acceptance Criteria
 - i. Invited users receive a notification saying they have been invited to a study group
 - ii. Invited users can accept or reject the invitation.
 - c. Test Cases: <https://github.com/peterwerner/citrix-capstone/blob/master/server-app/test/test.js>
6. Accept/Reject Invitation to a Study Session
 - a. As an invited user of a study session, I can accept or reject the invitation sent to me by a member of the study group so I can decide whether I want to be in the group or not
 - b. Acceptance Criteria
 - i. If the invite is accepted, the user is added to the preexisting video session using a group key
 - ii. If the invite is accepted, the user is led directly to the study session
 - iii. If the invite is rejected, nothing happens
7. Join a Study Session
 - a. As a student, I can join a previously created study session so I can study with my peers.
 - b. Acceptance Criteria
 - i. User is added to the preexisting video session using a group key
 - ii. User is led directly to the study session
 - c. Test Cases: <https://github.com/peterwerner/citrix-capstone/blob/master/server-app/test/test.js>
8. Leaving a Study Session
 - a. As the member of a study session, I can leave the study group so I can join a different group or leave the site completely.
 - b. Acceptance Criteria
 - i. User's id is removed from the group in the database
 - ii. User is removed from the study session
 - iii. User is led to his/her dashboard
 - iv. The study session persists without the user
9. Viewing Dashboard
 - a. As a student, I can view my dashboard so I can see and interact with the list of my classes and the create study session button.
 - b. Acceptance Criteria
 - i. User's class list is displayed on the screen with all relevant info (number of active study sessions)
 - ii. User is able to click and interact with the classes
 - iii. User can create a study session by clicking the create study session button
10. Viewing Course Page
 - a. As the member of a specific class, I can view the course's profile page

- b. Acceptance Criteria
 - i. User can see the active study sessions and their related topics
 - ii. User can view the list of course file uploads (the relevant files other users have uploaded)
 - iii. User can see the create study session button for the class

11. Adding Files to Course Page

- a. As the member of a specific class, I can upload files to the course page so I can post information I find relevant for other members of my class.
- b. Acceptance Criteria
 - i. Google Drive file appears in the file upload section of the course page
 - ii. File is sorted by most recent date added
 - iii. File topic tags are viewable below the icon

12. Viewing Chat Sidebar

- a. As a student, I can view a list of students who are online in an expandable/collapsible sidebar
- b. Acceptance Criteria
 - i. I can open the sidebar
 - ii. I can start/resume a chat with any of the students

13. Send a Chat Message

- a. As a student, I can send text messages to people in my study group(s)
- b. Acceptance Criteria
 - i. Message can be sent to server and received by the group(/person)
 - ii. Message is stored in chat history

14. Send a File/Photo/etc. via Message

- a. As a student, I can send a file to another student/group through chat
- b. Acceptance Criteria
 - i. File can be sent to server and received by the group(/person)
 - ii. File is stored in chat history

15. User Profile Management

- a. As a student, I can modify my profile information so I can control my account
- b. Acceptance Criteria
 - i. User can add/modify school(s) and class info
 - ii. School(s) and Class information are stored/updated in database
 - iii. User is led to the dashboard

Prototyping code

New code should be tested locally first, by running the server and client locally.

- Running the server locally - from the *server-app* directory, run *npm start*
- Running the client locally - from the *client-app* directory, run *npm start*

Deploying code

Code should be deployed once it is merged to master (see **GitHub Commits/Issues**).

- Deploying the server - from the *server-app* directory, run *npm run zip*, then upload the server archive to AWS Elastic Beanstalk
- Deploying the client - from the *client-app* directory, run *npm run build*, then run *npm run deploy*. (You must have the AWS CLI installed and have your AWS IAM account configured.)

Tests

Before merging to master and deploying, the automated tests should be run.

- Testing the server - from the *server-app* directory, run *npm run test*
- Testing the client - from the *client-app* directory, run *npm run test*

GitHub Commits/Issues

Committing new code:

1. Ensure the automated tests pass.
2. Commit to a new branch.
3. Open a pull request. Assign at least two other people to review it.
4. Once reviewed, merge with master locally.
5. Ensure the automated tests pass.
6. Merge the pull request to the master branch.
7. Deploy to AWS.

Issues are tracked on Waffle.io (integrated with GitHub issues). When you create a pull request for or merge in code resolving an issue, move the issue to the appropriate column on Waffle.io.

Appendix

Technologies

Client

- Static website stored on an AWS S3 bucket configured for static website hosting
- Front-end is generated by React components
- Communicates with API endpoints on the server (CORS - cross origin resource sharing)
 - Socket.IO for text chat - the client listens for data from and sends data to the server
 - WebRTC for video chat - the client requests a peer-to-peer connection to other clients from the server, sends data to and receives data from peer clients directly
- Communicates with Google APIs for public user profile information

Server

- Node JS server running on AWS Elastic Beanstalk (allowing automatic scaling up and down if we need more server instances)
- Responds to requests from the client (CORS - cross origin resource sharing)
 - Socket.IO for text chat, WebRTC for video chat - the server listens for data from the client, brokers connections between users, and sends data back to the client
 - Communicates with Google APIs for identity verification and access to private user information and files hosted on Google Drive

Database

- AWS DynamoDB non-relational database - accessed by the server to persist and retrieve data
 - User information (school IDs, course IDs, group IDs)
 - Group information (users IDs, file IDs)
 - School information (course IDs)
 - Course information (user IDs, group IDs)
- AWS S3 buckets
 - The client website is hosted in an S3 bucket configured for static website hosting
 - Files uploaded by users are stored in an S3 bucket